

Project Report on

Hybrid Firefly-Particle Swarm Optimisation for Feature Selection

Submitted in partial fulfillment of the requirements for the degree of

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in

INFORMATION TECHNOLOGY

by

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DECLARATION

We hereby *declare* that the Project Work Report entitled "***Hybrid Firefly-Particle Swarm Optimisation for Feature Selection***", which is being submitted to the **National Institute of Technology Karnataka, Surathkal**, for the award of the Degree of Bachelor of Technology in Information Technology, is a *bonafide report of the work carried out by us*. The material contained in this Project Report has not been submitted to any University or Institution for the award of any degree.

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CERTIFICATE

This is to *certify* that the Major Project Work Report entitled "***Hybrid Firefly-Particle Swarm Optimisation for Feature Selection***" submitted by

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as the record of the work carried out by them, is *accepted as the B.Tech. Project work report submission* in partial fulfillment of the requirement for the award of degree of Bachelor of Technology in Information Technology in the Department of Information Technology, NITK Surathkal.

Signature of Project Guide with date

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ABSTRACT

Feature selection is a crucial preprocessing step in machine learning, especially for high-dimensional datasets. It aims to identify the most informative subset of features that enhance classification accuracy while reducing model complexity. This paper presents a hybrid metaheuristic approach based on the Hybrid Firefly–Particle Swarm Optimization (HFPSO), which combines the exploratory power of the Firefly Algorithm (FA) with the exploitation ability of Particle Swarm Optimization (PSO) with a number of optimisations. We evaluate the modified HFPSO on three benchmark datasets: Arcene, Dorothea, and Madelon, using K-Nearest Neighbor (KNN) as a wrapper evaluator. Experimental results demonstrate that the modified HFPSO effectively selects compact feature subsets with competitive predictive performance compared to the baseline HFPSO.

Keywords— Feature selection, Firefly algorithm, Particle swarm optimization, Hybrid metaheuristics

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CHAPTER 1

INTRODUCTION

1.1 Overview

Feature selection is a crucial preprocessing step in machine learning, especially for high-dimensional datasets. It aims to identify the most informative subset of features that enhance classification accuracy while reducing model complexity. With increasingly large and complex datasets, the need for advanced feature selection technologies are required more than ever.

Any research into particle swarm optimisation style techniques must also improve upon the existing knowledge of the algorithmic behaviour of particles. By modifying the equations and calculations present, it is possible to obtain preferred searching from the particles themselves finding optimal results with less compute and over fewer epochs of simulation.

1.2 Motivation

Particle Swarm Optimisation techniques are dependant on the conversion of local search results into overall global search results. That is, the local minute movements of the particles must be interpreted to find the global optimal solution. Ragab et al. 2024 [3] uses a Hybrid Firefly-Particle Swarm Optimisation to balance the efficacy of these two goals.

The Firefly optimisation algorithms are preferred for local search whereas the Particle Swarm Optimisation algorithm is preferred for local search. By comparing the fitness of particles with the global optimum, a corresponding algorithm is selected and used.

We find that there are a number of optimisations that can be added to the code to obtain a faster result over fewer epochs.

CHAPTER 2

LITERATURE REVIEW

2.1 Background and Related Works

There exists a large variety of particle swarm based optimisation techniques based on techniques inspired by nature and genetic algorithms. An example of these is the Firefly algorithm described by Yang et al. 2010 [2]. It involves updating the present location based on the global best location weighted down by an exponential factor based on distance. This one in particular does well on local searching.

Some others such as the Particle Swarm Optimisation algorithm are based on purely mathematical principles. It involves updating the present location based on a social factor that accounts for the global best and a cognitive factor that accounts for the particle's historical best. Due to this method, it is better at a long-range global search.

2.2 Outcome of Literature Review

The Hybrid Firefly-Particle Swarm Optimisation algorithm introduced by Ragab et al. 2024 [3] considers using the Firefly algorithm's strength in local search and the Particle Swarm Optimisation's strength in global search to construct a combined algorithm that implements either depending on the particle's present fitness and the global best. This allows the algorithm to explicitly choose between exploration and exploitation in addition to the implicitly defined decisions present within the algorithms themselves.

The choosing of the features is based on a wrapper method wherein a threshold on the position is used to choose the features. This can be modified to allow the user to decide the number of features being selected for by choosing the top k largest features. In addition, the remaining features could be reduced down allowing for the information present in them to be used as well at the cost of increased parametrisation and computational costs.

Further, while the Particle Swarm Optimisation algorithm includes a linearly reducing weight that controls the prevalence of exploration vs. exploitation, the Firefly algorithm has no such feature. Thus we can explicitly define a weight that reduces either linearly or exponentially that can control these factors.

Other features that directly reduce the limitations of the Hybrid Firefly-Particle Swarm Optimisation algorithm are the introduction of the global and personal worst positions. These allow the particles to avoid areas similar to the use of global and personal bests to promote areas.

Along with this, the choice between Particle Swarm Optimisation and Firefly is based instead on the change in the fitness rather than the absolute value as that is more representative of the value of the particles position. In addition, to prevent pooling, particles that attain the same feature set as the global best are reset. This promotes further exploration beyond what is already known.

2.3 Problem Statement

To develop an oracle-based feature selection algorithm based on a number of simulated particles whose positions relate to the information gain and that utilises Firefly and Particle Swarm Optimisation in tandem to promote local and global search when needed.

2.4 Objectives of the Project

- (1) Conversion of the feature selection strategy to be based on top k features rather than a threshold-based strategy.
- (2) Utilising changes in fitness rather than absolute fitness to choose between the two algorithms.
- (3) Introducing global worst and personal worst into the Firefly and Particle Swarm Optimisation algorithms.
- (4) Combining unselected features into a reduced feature set.
- (5) Adding individual particle weights as well as resetting particles when they attain global best.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Particle Swarm Optimisation Algorithm

Each particle i has a position $\mathbf{x}_i$ and velocity $\mathbf{v}_i$. These are randomly initialised uniformly over a range 0 to 1 for every feature in the dataset. At each step of the simulation, the position and velocity are updated according to the equations:

$$\mathbf{v}_i(t+1) = w \mathbf{v}_i(t) + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i) \quad (1)$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \quad (2)$$

where, w is the inertia weight and c_1 and c_2 are learning factors respectively. Here, r_1 and r_2 are random numbers in the range 0 to 1.

The positions and velocities are then capped to be in the range 0 to 1 for all the features in the dataset. Finally, the weights are updated according to the equation:

$$w = w_{\text{initial}} - (w_{\text{final}} - w_{\text{initial}}) * (l_{\text{curr}} / l_{\text{max}}) \quad (3)$$

where, w_{initial} and w_{final} are the initial and final weights and l_{curr} and l_{max} are the current and maximum iterations.

Algorithm 2 PSO for Feature Selection

```

1: Initialize particle positions  $\mathbf{x}_i$ , velocities  $\mathbf{v}_i$ 
2: for each iteration do
3:   for each particle  $i$  do
4:     Update velocity and position using (1)–(2)
5:     Evaluate KNN accuracy; update  $\mathbf{p}_i$  and  $\mathbf{g}$ 
6:   end for
7: end for
8: Return global best subset

```

Algorithm 1: Particle Swarm Optimisation

3.2 Firefly Algorithm

In the firefly algorithm, each particle is an individual firefly. It's brightness is measured by its fitness and fireflies update their position based on the brightness of the other fireflies. The positions are updated based on the equation:

$$\beta(r) = \beta_0 e^{-\gamma ||g-x_i||^2} \quad (4)$$

$$x_i(t+1) = x_i(t) + \beta(g-x_i) + \alpha(r_3 - 0.5) \quad (5)$$

where α , β , γ are learning factors and r_3 is a random number in the range 0 to 1. The algorithm can be written out as follows:

Algorithm 1 Firefly Algorithm for Feature Selection

```

1: Initialize  $n$  fireflies (feature subsets)
2: for each iteration do
3:   for each  $i, j$  do
4:     if  $f(x_j) < f(x_i)$  then
5:       Move  $x_i \leftarrow x_i + \beta_0 e^{-\gamma r_{ij}^2} (x_j - x_i) + \alpha(\text{rand} - 0.5)$ 
6:     end if
7:   end for
8: end for
9: Return best subset

```

Algorithm 2: Firefly Algorithm

3.3 Hybrid Firefly-Particle Swarm Optimisation

Here at each epoch, each particle's fitness is measured. If the fitness is above the global best fitness from the previous epoch, the particle's position is updated using Firefly algorithm, otherwise it is updated using the Particle Swarm optimisation algorithm. The overall flowchart is as follows:

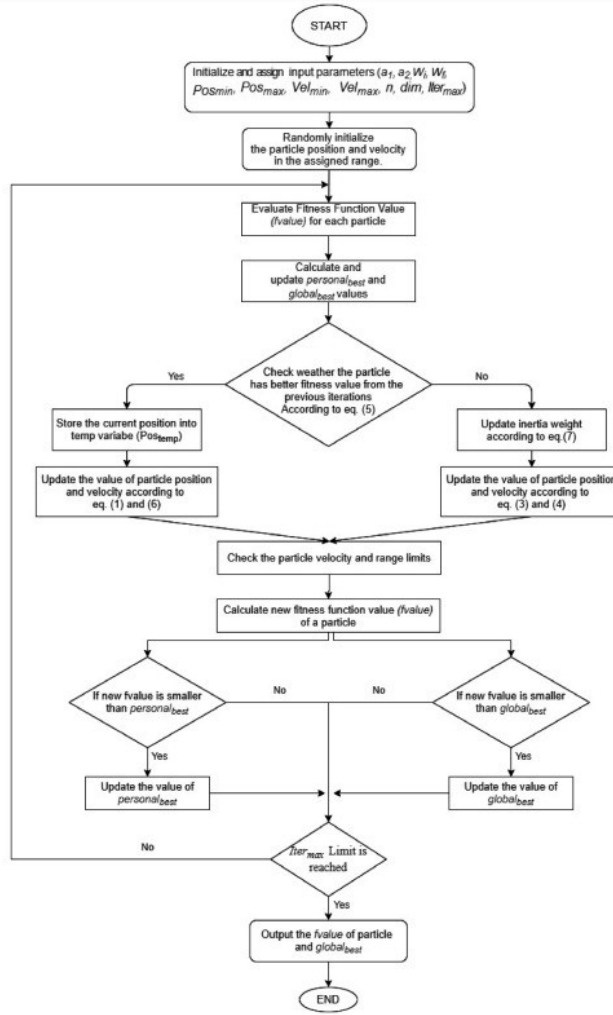


Image 1: Hybrid Firefly-Particle Swarm Optimisation Flowchart

For the modified Hybrid Firefly-Particle Swarm Optimisation algorithm, the updating instead based on the change in fitness. If the change in fitness is greater than that of the global best, then the firefly algorithm is used, otherwise firefly algorithm is used. Further, particles that attain the same features i.e. similar position to the global best are repositioned randomly to begin the exploration phase again.

Each particle is updated according to the modified versions of equations (1) and (2) as follows:

$$\mathbf{v}_i(t+1) = w \mathbf{v}_i(t) + (1-T_i)c_1r_1[(\mathbf{p}_i-\mathbf{x}_i) - (*\mathbf{p}_i-\mathbf{x}_i)] + T_ic_2r_2[(\mathbf{g}-\mathbf{x}_i) - (*\mathbf{g}-\mathbf{x}_i)] \quad (6)$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \quad (7)$$

Further, equations (4) and (5) are modified as follows:

$$\beta_1(r) = T_i \beta_0^{-\gamma ||g-x_i||^2} \quad (8)$$

$$\beta_2(r) = T_i \beta_0^{-\gamma ||g-x_i||^2} \quad (9)$$

$$x_i(t+1) = (1-T_i)x_i(t) + \beta_1(g-x_i) - \beta_2(*g-x_i) + \alpha(r_3 - 0.5) \quad (10)$$

where, *p and *g are the worst personal and global positions and T_i is the temperature of the particle. This temperature is updated according to the equation:

$$T_i(t+1) = (1-p)T_i(t) \quad (11)$$

where p is a predetermined factor. This follows the following algorithm where τ represents the change in fitness for the global best:

Algorithm 3 Hybrid Firefly–PSO (HFPSO)

```

1: Initialize population, velocities, fitness values
2: for each iteration do
3:   for each particle  $i$  do
4:     if change in fitness  $> \tau$  then
5:       Perform FA local update
6:     else
7:       Perform PSO global update
8:     end if
9:     if duplicate global fitness then
10:      Reinitialize particle for diversity
11:    end if
12:    Evaluate KNN-based fitness and update bests
13:  end for
14: end for
15: Return best feature subset

```

Algorithm 3: Hybrid Firefly-Particle Swarm Optimisation

Lastly, unused features are reduced into a predetermined number of dimensions by using Principal Component Analysis (PCA).

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Results

We run the described algorithm on three datasets: Arcene, Dorothea and Madalon. These datasets were chosen for their high feature counts with them also having several non-informative features that don't need to be selected for high accuracy. Running both the unmodified and modified Hybrid Firefly-Particle Swarm Optimisation Algorithms on the datasets we get the following results:

Dataset	Features	Accuracy	Precision	F1-score
Arcene	4998	0.830	0.835	0.831
Dorothea	50066	0.903	0.815	0.857
Madelon	252	0.575	0.574	0.573

Dataset	Features	Accuracy	Precision	F1-score
Arcene (Enhanced)	2412	0.790	0.795	0.791
Dorothea (Enhanced)	197	0.902	0.810	0.902
Madelon (Enhanced)	183	0.590	0.595	0.595

Table 1: Results from modified and unmodified algorithms

We find that the enhanced algorithm finds much better results with fewer features than the unmodified algorithm. In addition we find that the algorithm obtains its results in fewer epochs than the unmodified algorithm as can be seen in the following graphs.

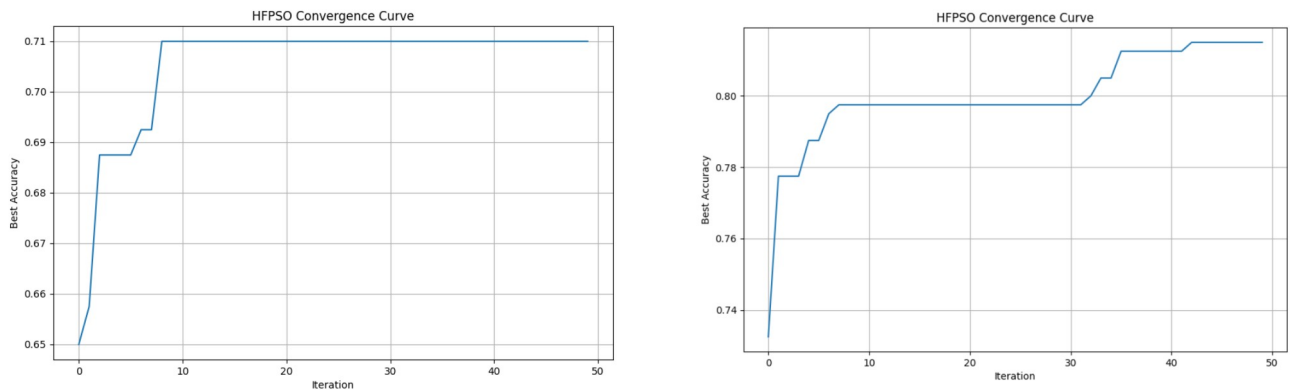


Image 2: Best accuracies per epoch of the modified and unmodified algorithm

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

We describe a modified Hybrid Firefly-Particle Swarm algorithm that is able to obtain faster results. Further, these results utilise a fewer number of features to obtain accuracies that rival if not surpass the accuracies obtained by the algorithm. These results are achieved by the addition of features that promote rapid exploration as well as prevent bottlenecks in movement of particles within the search space.

Future avenues of research might find use for the positions as fractional weights thereby treating them as weights for the features themselves. This can help training time but mainly benefits model accuracies. The main issue in this regard is balancing the benefits with the harmful effects of causing some features to be ignored.

Another interesting research subject might be the use of advanced capping algorithms for the position and velocity instead of manually limiting to the range of 0 to 1. These algorithms may help prevent unnecessary loss in precision caused by the manual clipping attempts.

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